



Calculus 1M – Final A (Summer 2021)

INSTRUCTIONS: Write Your Answers to each problem **on the SAME PAGE** in the blank space provided. **NO Electronic Devices or Notes Allowed.**

There are two parts of the exam:

- **Part 1:** Write the complete solution like in the Written HW.
- **Part 2:** Fill the answer box like in the Webwork HW.

Your Name: _____

Student ID: _____

Good luck! 😊

Part 1: PROBLEMS 1-6, Show All Work For a Full Credit.

1. **(10 points)** Find

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x+2} \right)^x.$$

2. **(10 points)** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} ax^2 + b & \text{if } x \leq 1 \\ \frac{4}{\sqrt{x}} + 1 & \text{if } x > 1 \end{cases}.$$

Find the values of the real numbers a and b such that the function is differentiable in its domain.

Hint: A relation between differentiability and continuity at the point could be used to find the second parameter.

3. **(10 points)** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function.

- a) Find the *Absolute Minimum* and *Absolute Maximum* of the function on $[-2, 1]$.
- b) Find all the *Local Extrema* of this function on $\mathbb{R}$.

For Calculus 1M ONLY:

$$f(x) = x^3 - \frac{3}{2}x^2 - 6|x| + 10$$

4. **(10 points)** Find the volume of the solid generated by rotating the region bounded by the given curves around the y -axis.

For Calculus 1M ONLY: $y = x^2, y = \sqrt{8x}$.

5. **(10 points)** Test the given problems for the convergence or divergence. Specify the test used. If it converges, find the value.

For Calculus 1M ONLY:

a) $\int_1^2 \frac{1}{x \ln x} dx.$

b) $\sum_{n=0}^{\infty} \frac{(-2)^n}{5^{n+1}}.$

6. **(10 points)**

- a) Write the Taylor polynomial of degree 3 for the function $f(x) = 32\sqrt{x}$ around $a = 4$.
- b) Use the second degree polynomial (from the part above) to approximate $\sqrt{2}$.
- c) Estimate the error of an approximation.

Part 2: PROBLEMS 7-12 Fill the Answer Box. Explanations are NOT required and NOT graded.

7. (8 points) Find the following limits:

$\lim_{x \rightarrow 0^+} \frac{ \sin x }{x} =$	$\lim_{x \rightarrow 0^-} \frac{ \sin x }{x} =$
$\lim_{t \rightarrow 0} \frac{\tan 3t}{5t} =$	$\lim_{x \rightarrow -\infty} \frac{3 + \sqrt[3]{x} - 5x}{-4 + 2x + x^{2/3}} =$

8. (10 points) Find $\frac{dy}{dx}$ using explicit or implicit differentiation.

If $y = \left(\frac{e^x - x}{1 + \sin x} \right)^3$ then $\frac{dy}{dx} =$
If $xy^2 = \ln x$ then $\frac{dy}{dx} =$

9. (5 points) Let

$$\int_1^x f(t) dt = x \cos 2x.$$

Find $f\left(\frac{\pi}{2}\right) =$

10. (9 points) Let a rational function is $f(x) = \frac{4x^2 - 9x + 3}{x(x-1)^2}$.

Write the partial fraction decomposition of $f(x)$.

$f(x) =$

Evaluate $\int f(x) dx =$

11. **(4 points)** Suppose that $\sum_{n=0}^{\infty} c_n x^n$ converges when $x = -3$ and diverges when $x = 6$. Answer “Converges” or “Diverges” for the following series.

$\sum_{n=0}^{\infty} c_n$	$\sum_{n=0}^{\infty} c_n 9^n$
$\sum_{n=0}^{\infty} c_n (-8)^n$	$\sum_{n=0}^{\infty} (-1)^n c_n 2^n$

12. **(4 points)** Let the series is

$$\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{2^n (n+3)}$$

Find the <i>Radius of Convergence</i> $R =$
Find the <i>Interval of Convergence</i>